



JAIPUR NATIONAL UNIVERSITY

End Semester/ Year Examination December -2025

Name of School:	School of Computer and Systems Sciences
Name of Programme / Course:	General Elective
Semester/Year:	V / III
Paper Code:	SCSSAGE501T22
Title of Paper:	Introduction to Software Solution Building
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

1. This question paper consists of two sections, Section-A and Section-B.
 2. Section-A consists of seven questions and
 3. Section-B consists of six questions.
- Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question

Question Number	SECTION - A	Marks Allotted	Bloom Level	Course Outcome
Q.1	Define Software Process. Explain various Software Engineering Paradigms with suitable examples.	4	Understand	CO1
Q.2	Differentiate between the Waterfall Model and Incremental Model. Discuss their advantages and limitations.	4	Analyze	CO1
Q.3	Explain the Spiral Model and Prototyping Model. How do they support risk management and early feedback?	4	Understand	CO2
Q.4	Describe the Verification and Validation process in software	4	Apply	CO2

engineering. Why are both essential?

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|-----|--|---|------------|-----|
| Q.5 | Define Functional and Non-Functional Requirements. Explain the Requirement Engineering Process. | 4 | Understand | CO3 |
| Q.6 | Explain Feasibility Study in software development. What are the different types of feasibility analyses performed? | 4 | Analyze | CO3 |
| Q.7 | Write short notes on:
(a) System Engineering Hierarchy,
(b) Computer-Based Systems. | 4 | Remember | CO3 |

SECTION - B

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|-----|---|----|------------|-----|
| Q.1 | Explain Software Document Analysis and Modeling. Discuss data, functional, and behavioral models with examples. | 10 | Understand | CO4 |
| Q.2 | What is Modular Design? Explain Design Concepts, Principles, and Heuristics in detail. | 10 | Understand | CO4 |
| Q.3 | Explain Architectural Design and Data Design. Discuss Transform Mapping and Transaction Mapping with suitable examples. | 10 | Apply | CO5 |
| Q.4 | Discuss User Interface Design Principles. How do User Interface Heuristics improve usability? | 10 | Analyze | CO5 |

~~Q.5~~

Describe Real-Time Software Design. Explain Data Acquisition Systems and Monitoring & Control Systems.

10

Understand

CO6

~~Q.6~~

Write short notes on:

10

Remember

CO6

(a) Software Prototyping Techniques,

(b) Rapid User Interface Prototyping.